

MANAGUA INTL, Nicaragua

WMO: 787410

Lat: 12.141N

Lon: 86.168W

Elev: 194

StdP: 14.59

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.0	69.7	58.6	74.0	82.3	60.4	79.0	83.4	22.0	88.1	19.9	87.7	1.4	110	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.9	96.8	75.9	95.3	75.6	94.7	75.6	79.9	88.6	79.3	88.1	79.0	87.8	10.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.4	83.1	77.2	142.4	82.9	76.8	140.6	82.7	43.5	89.1	43.0	88.8	42.5	88.5	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.6	15.6	DB	63.8	98.6	3.3	1.0	61.5	99.3	59.5	99.9	57.7	100.4	55.3	101.1
			WB	59.8	81.6	2.5	1.5	58.0	82.7	56.5	83.6	55.1	84.4	53.3	85.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	80.4	81.6	83.3	85.4	84.9	82.7	82.4	82.8	82.2	81.3	80.6	80.4	(6)
	DBStd	2.44	1.78	1.84	2.11	1.87	2.47	1.87	1.73	1.63	1.70	1.86	1.73	1.87	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11802	944	884	1032	1061	1080	981	1004	1017	967	971	918	943	(10)
	CDD65	6327	479	464	567	611	615	531	539	552	517	506	468	478	(11)
	CDH74	67347	4825	5030	6883	8039	7475	5431	5356	5592	4964	4632	4439	4681	(12)
	CDH80	29691	2063	2304	3450	4271	3656	2133	2099	2339	1971	1698	1743	1964	(13)
(14) Wind	WSAvg	6.5	8.2	9.0	9.5	8.7	6.7	5.2	6.0	5.3	4.4	4.0	4.6	6.6	(14)
(15) Precipitation	PrecAvg	43.6	0.2	0.0	0.0	0.2	4.4	8.1	5.2	6.1	8.1	8.4	2.2	0.4	(15)
	PrecMax	68.6	0.7	0.1	0.3	1.3	8.9	16.6	10.5	14.3	14.4	16.9	8.0	1.5	(16)
	PrecMin	26.4	0.0	0.0	0.0	0.0	0.1	3.2	2.0	1.6	2.8	2.5	0.2	0.0	(17)
	PrecStd	11.5	0.2	0.0	0.1	0.4	2.9	3.3	2.4	3.0	3.2	4.2	2.0	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	95.0	97.0	98.6	98.4	94.8	93.5	95.0	93.6	91.8	91.7	93.0	(19)
		MCWB	74.1	75.2	75.7	76.1	76.4	77.6	77.0	78.0	78.2	78.2	76.6	75.7	(20)
	2%	DB	91.5	93.2	95.3	96.9	96.7	92.9	91.7	93.1	91.9	91.1	90.0	91.4	(21)
		MCWB	73.6	74.2	74.9	75.7	76.2	77.9	77.1	77.7	78.1	78.0	76.2	74.9	(22)
	5%	DB	90.0	91.7	93.6	95.4	95.1	91.3	91.0	91.5	91.2	89.6	89.5	89.8	(23)
		MCWB	73.2	73.6	74.0	75.3	76.2	77.8	77.1	77.7	78.0	77.6	76.0	74.5	(24)
	10%	DB	89.3	90.4	93.0	94.7	93.3	89.5	89.4	89.8	89.5	88.0	88.0	88.7	(25)
		MCWB	73.0	73.2	73.7	75.1	76.2	77.6	77.0	77.4	77.7	77.7	75.9	74.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.6	78.0	78.5	79.7	80.1	80.1	79.8	80.0	80.1	80.1	79.2	78.0	(27)
		MCDB	88.2	89.7	92.0	90.3	89.8	88.6	87.8	89.3	88.4	87.7	87.4	87.3	(28)
	2%	WB	75.5	76.2	77.1	78.4	79.2	79.5	79.0	79.2	79.5	79.2	78.5	76.8	(29)
		MCDB	87.3	88.5	90.3	90.1	88.6	87.9	87.4	88.5	87.9	87.0	86.7	86.9	(30)
	5%	WB	74.7	75.2	76.1	77.5	78.6	79.0	78.3	78.8	79.0	78.8	77.7	76.0	(31)
		MCDB	86.3	87.5	89.2	89.2	88.0	87.3	86.8	88.1	87.5	86.6	86.0	86.1	(32)
	10%	WB	73.9	74.3	75.1	76.8	77.9	78.4	77.8	78.1	78.4	78.2	77.0	75.2	(33)
		MCDB	85.2	86.7	88.2	88.6	87.3	86.4	86.0	87.0	86.7	85.6	85.2	85.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	19.7	20.5	19.9	17.0	14.5	14.9	16.0	15.4	14.2	16.4	18.2	(35)
		MCDBR	19.8	20.7	21.1	20.5	19.4	16.4	16.7	17.5	17.4	16.3	17.7	19.0	(36)
	5% WB	MCWBR	6.5	6.6	6.3	5.8	5.4	5.7	5.7	6.0	6.2	6.2	6.8	6.5	(37)
		MCWBR	18.7	19.9	20.2	19.3	16.5	14.7	14.9	16.2	15.6	14.7	15.6	17.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.366	0.371	0.380	0.453	0.499	0.446	0.448	0.443	0.427	0.405	0.377	0.371	(40)
		taud	2.555	2.538	2.501	2.277	2.173	2.361	2.351	2.360	2.403	2.489	2.572	2.569	(41)
	Ebn at Noon	Ebn at Noon	298	299	297	273	256	267	267	272	279	285	292	293	(42)
		Edh at Noon	32	34	36	44	48	39	40	40	38	35	31	31	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1680	1872	2057	1994	1711	1628	1642	1731	1689	1569	1613	1596	(44)
	RadStd	RadStd	59	76	67	88	98	79	90	74	119	141	73	71	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+0.61	+1.03	N/A	+0.29	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page